

JT\_DEV\_B44\_1.3

## API in TypeScript with Bun & SQLite

### 1 Initial situation

- 1.1 In your previous project, you developed a Todo List application using React. You used an existing API to store and retrieve your tasks. Now, you will build your own backend to replace that API.
- 1.2 You will use [Bun](#), a fast all-in-one JavaScript runtime (yes, one more runtime...), and [SQLite](#), a lightweight SQL database engine. Bun has [built-in support for SQLite](#), making it very easy to set up a database without a complex configuration.
- 1.3 The objective is to recreate the endpoints used by your React application so that you can switch from the [Todo API](#) to your own local API seamlessly. You should not change your React code except for the API URL.

### 2 Skills from the ICT training plan worked on

Skills ranked by importance of implementation in the project: [consult the training plan](#).

- 2.1 g5 : Implémenter les applications et les interfaces selon le concept en respectant les exigences de sécurité.
- 2.2 c2 : Mettre en œuvre des modèles de données dans un dispositif de stockage de données numériques.
- 2.3 g1 : Analyser et documenter les besoins en matière d'applications et d'interfaces.
- 2.4 a4 : Planifier les projets ICT et les tâches selon un modèle de procédure.
- 2.5 a6 : Vérifier l'avancement des projets ICT et des tâches et en faire état selon le modèle de procédure.
- 2.6 g6 : Vérifier la qualité et la sécurité des applications et des interfaces.

### 3 Deliverables

- 3.1 GitHub repository containing your backend project.
- 3.2 Your previous React project updated to use your new local API.

### 4 Andragogic conditions

- 4.1 Work is carried out independently, on an individual basis.
- 4.2 You are allowed to help each other, share ideas and discuss solutions, but each trainee must provide their own implementation with their own code.
- 4.3 Apprentices have access to all the resources they need, particularly on the web.
- 4.4 Don't hesitate to ask AI for explanations, but avoid asking it for answers. The more you ask for answers, the less you learn.
- 4.5 **Maximum allowed time for the project: 4 full days (7.5 hours per day).** You must manage your time independently. On project beginning, inform your trainer of the start date and time.

### 5 Steps

- 5.1 Install Bun on your system with the official convenience script. Check instructions here: <https://bun.com/docs/installation>



- 5.2 Create a new GitHub Project from [the bun API template](#) to your account (Use the «...» then «Make a copy» menu). Select your account to host the Project on and name the project with your GitHub pseudo, for example `bun-todo-api-fuzoh`. **Do not forget to tick the «Draft issues will be copied if selected» checkbox!**
- 5.3 Then, you just have to implement all the features requested in the GitHub Project. **For each task, take time to read them attentively and complete the `estimate` field with the number of hours you expect to work on it. Before merging your code check that all acceptance criteria are met**, take review feedback into account. Then, complete the `final` field with the actual time spent on the task.

## 6 Resources

- 6.1 Bun installation : <https://bun.com/docs/installation>
- 6.2 Start a minimal Bun project : <https://bun.com/docs/quickstart>
- 6.3 Managing packages with bun : <https://bun.com/docs/pm/cli/install>
- 6.4 Bun router : <https://bun.com/docs/runtime/http/server>
- 6.5 Bun SQLite support : <https://bun.com/docs/runtime/sqlite>
- 6.6 SQL basics <https://sql.sh/>

## 7 Evaluation criteria

- 7.1 The API mimics the behaviour of the original API (same JSON structure) for the `/todos` endpoint.
- 7.2 Data is persisted in the SQLite database.
- 7.3 The React application works fully with the new backend.
- 7.4 The code is clean, typed (TypeScript), and follows conventions.
- 7.5 The code is well organised (KISS, separation of concerns)
- 7.6 Commits follow the conventional commits standard.
- 7.7 Code is merged through PR with close keywords
- 7.8 Errors are well managed (comprehensive information displayed to the client)
- 7.9 A small README.md gives the important information to work with the repo

